Chapter 5 exercises

DS05-E01 • Traversals and height

The root is 8. Its left child 3 has children 1 and 6. Its right child 12 has left child 10. Find preorder, inorder, postorder, level order, tree height, every node's depth, and maximum level width. State DFS and BFS auxiliary-space bounds.

DS05-E02 • BST insertion and two-child deletion

Insert 11 into the BST in DS05-E01, then delete 8 using its successor. Draw the result and identify the physically removed node position. Explain the semantic issue if external handles identify nodes and deletion moves values.

DS05-E03 • Heap updates

Start with min-heap `[2,5,4,9,7,8]`. Insert 1, then delete-min, listing every swap. Do not replace heap operations with sorting the whole array.

DS05-E04 • Prove linear heap construction

Explain why "n sift-downs, each O(log n), hence Θ(n log n)" is not a tight-bound proof. Prove Θ(n) by summing costs over heights. State the worst-case cost of construction by individual insertions.

DS05-E05 • From ordered sources to a balanced tree

Repeatable ordered sources are [2,7,17],[5,7,13],[11,19], with value 1 per record. Aggregate equal keys into occurrence counts. Design construction without repeated ordinary BST insertion; show every key/value, parent, and height. Separate merging and construction costs when source count k grows. Trace successor(7) and the maximum key’ s successor.

DS05-E06 • Branch counts and the empty tree

A nonempty binary tree has 55 nodes, of which 22 have exactly one child. Find its leaf count and two-child count, deriving the result by counting edges. Does it apply to the empty tree?

DS05-E07 • Reconstruct and test uniqueness

Preorder is [8,3,1,6,12,10] and inorder is [1,3,6,8,10,12]. Show each root and interval split, draw the tree, and give postorder. Give two different trees with preorder [A,B] and postorder [B,A].

DS05-E08 • First-child next-sibling encoding

The ordered children of A are B,C,D; B has E,F; C has G; D has H. List first-child and next-sibling for every node. Recover the children of A and B. When B.right points to C, is C a child of B in the original tree?

DS05-E09 • Shapes, labels, and BSTs

Derive the recurrence by splitting on the root's left subtree size and compute C_0 through C_4. How many BSTs use three fixed distinct ordered keys? How many general binary trees use three distinct labels with arbitrary placement? Explain the distinction.

DS05-E10 • Complete-tree indices and height

For a complete tree with n=10 and zero-based indices, give node 4's children, node 8's parent, all leaf indices, and tree height. How does the height change when counted in levels starting at one?

DS05-E11 • Bottom-up heap construction

Apply min-heap sift-down to [9,5,6,2,3,8,7] for i=2 down to 0. Show comparisons, each swap, and the complete array; draw the final tree. Are both child subtrees heaps before processing i=0?

DS05-E12 • Expression tree and stack evaluation

Write preorder and postorder for `(3+5)×(9−6)`. Evaluate postorder with a stack, showing the stack before and after each operator. Explain the subtraction operand order.

DS05-E13 • Catalan counting

Compute C_4 from the left-subtree recurrence and explain why the path-reflection argument gives $C_4 = \binom{8}{4} - \binom{8}{3}$. How many BST shapes use four fixed distinct keys?

References and attribution

Lin Cing, handwritten data structures notes; ISO C++ draft priority_queue.

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https://github.com/xinhi6716/computer-science-self-study

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